

Figure S1 . Distribution of 1120 SNPs at 5 GWAS loci

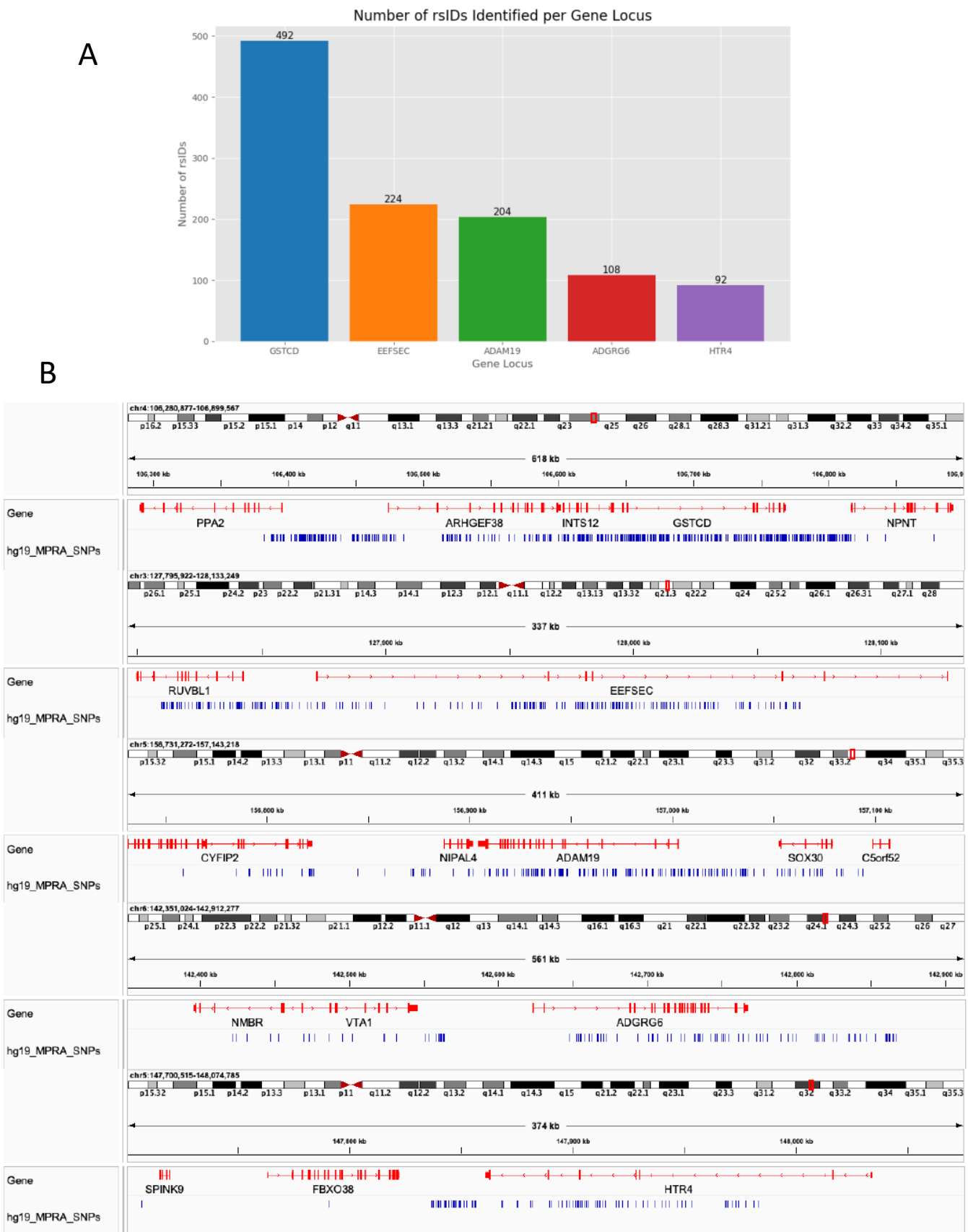


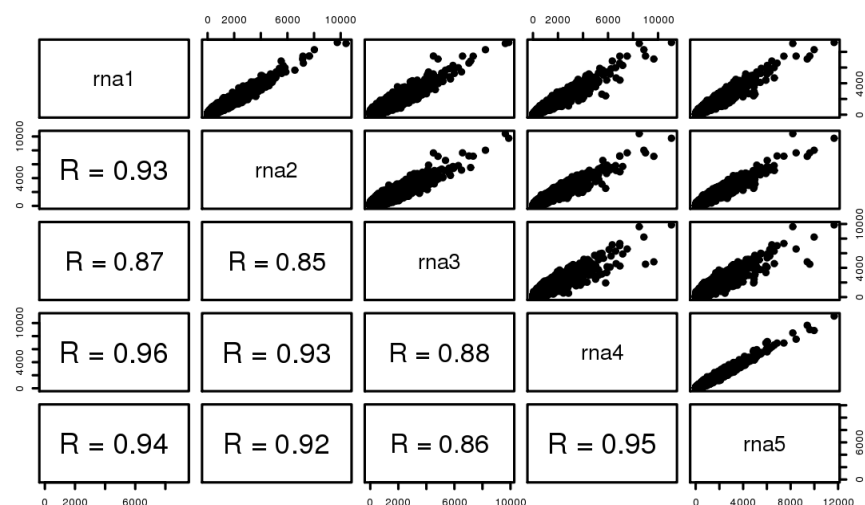
Figure S2. MPRA sequencing reads in 16HBE cells (five replicates)

Criteria	Plasmid	DNA1	DNA2	DNA3	DNA4	DNA5
# reads	19,234,197	36,049,298	34,388,119	23,004,723	22,081,168	23,893,532
# reads in the library	14,555,680 (75.7%)	27,308,920 (75.8%)	26,108,101 (75.9%)	17,583,081 (76.4%)	16,865,890 (76.4%)	18,090,779 (75.7%)
# tag> 0	87,033 (94.2%)	87,940 (95.2%)	87,846 (95.1%)	87,295 (94.5%)	87,240 (94.4%)	87,423 (94.6%)

Criteria	RNA1	RNA2	RNA3	RNA4	RNA5
# reads	6,512,429	7,482,838	7,873,726	7,046,396	8,384,238
# reads in the library	5,016,173 (77.0%)	5,757,873 (76.9%)	6,049,221 (76.8%)	5,423,297 (77.0%)	6,453,093 (77.0%)
# tag> 0	83,472 (90.3%)	79,495 (86.0%)	70,349 (76.1%)	84,165 (91.0%)	82,014 (88.8%)

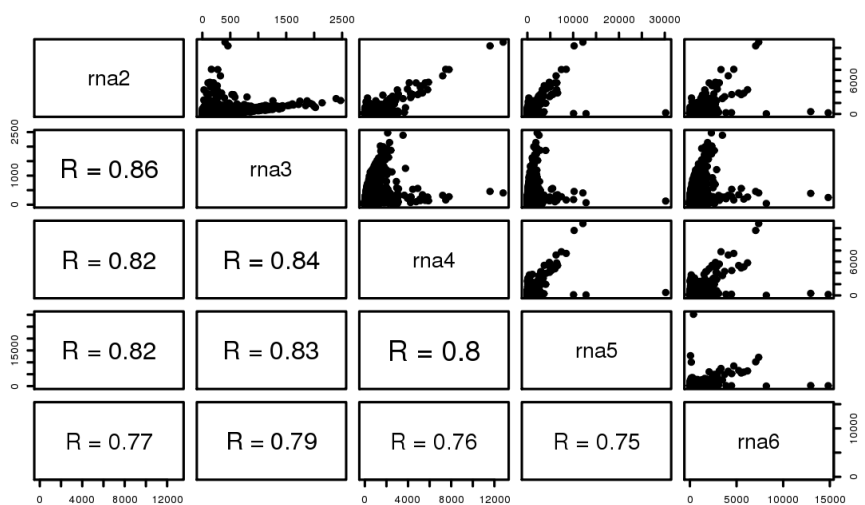
Figure S3. Correlation of RNA sequencing results in multiple replicates in three cell types.

A



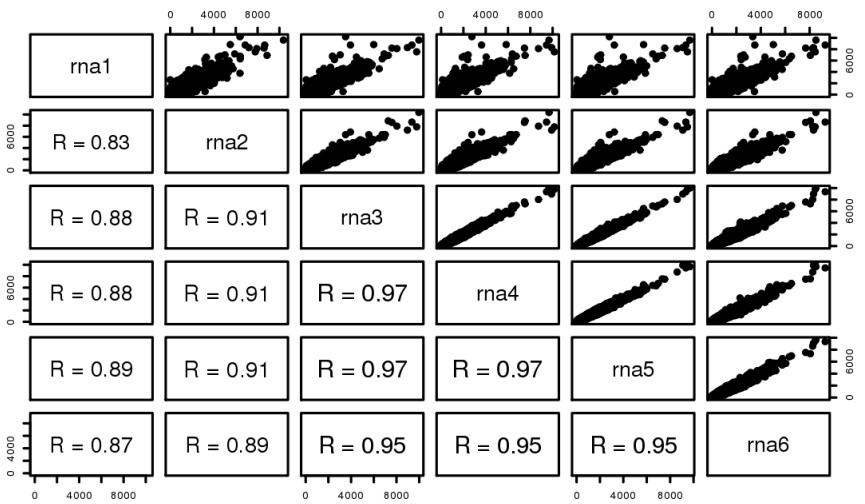
16HBE

B



HUVEC

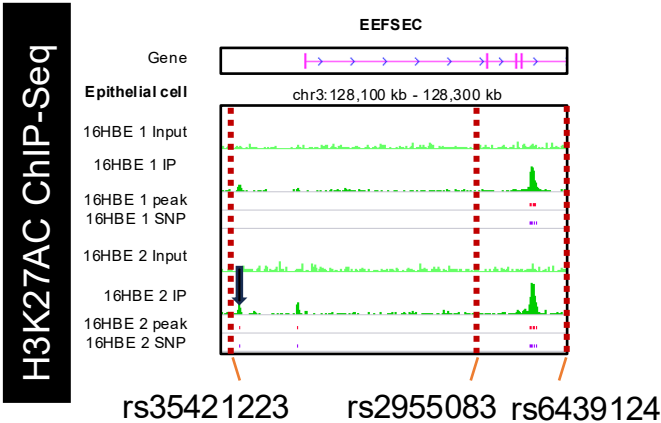
C



MRC5

Figure S4. Measurements on the open chromatin near the MPRA significant variants

A



B

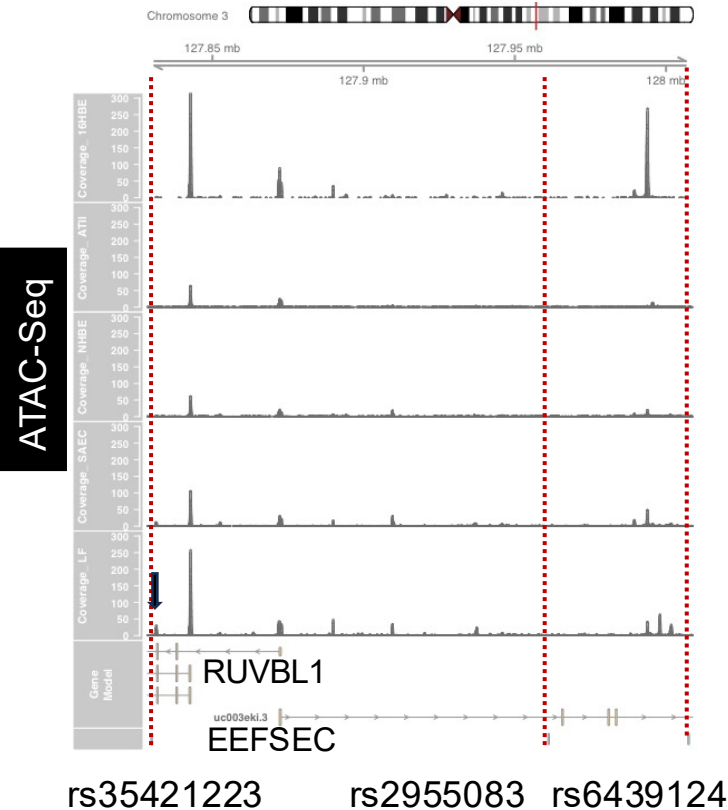
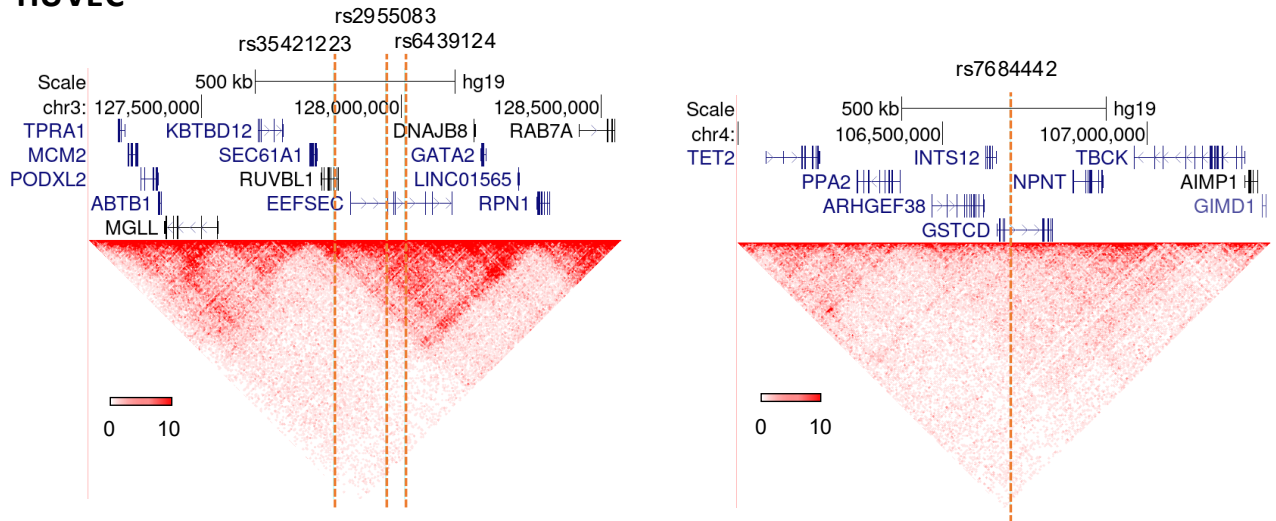


Figure S5. Public HiC data near MPRA significant variants in HUVEC and IMR90 cells

A

HUVEC



B

IMR90

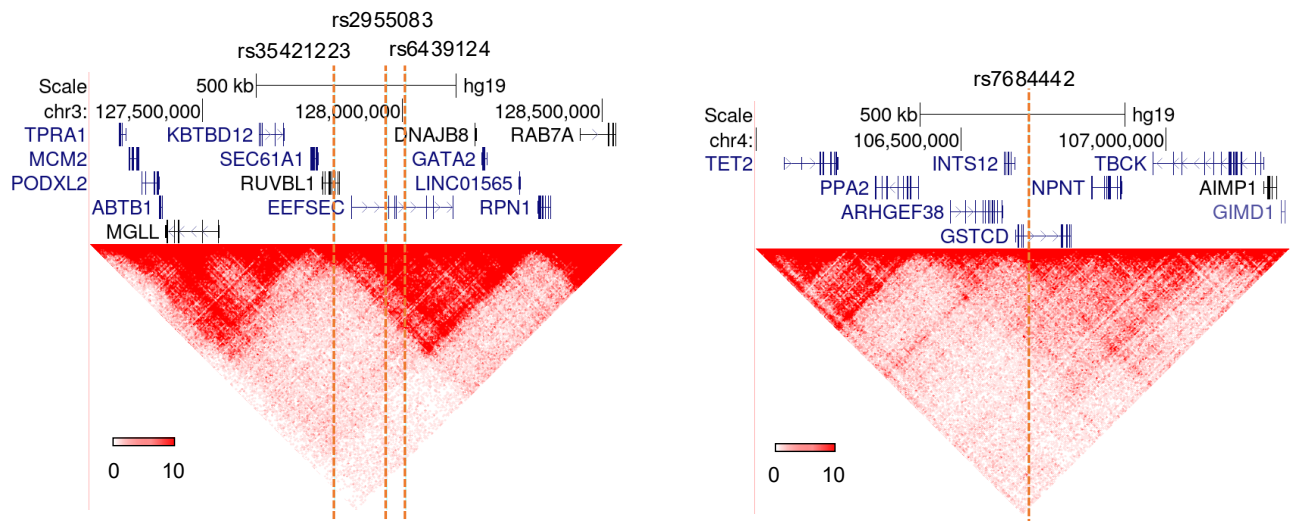


Figure S6. Gene expression measurements in 16HBE cells after CRISPRi targeting four MPRA significant variants with three replicates and two gRNAs for each SNP.

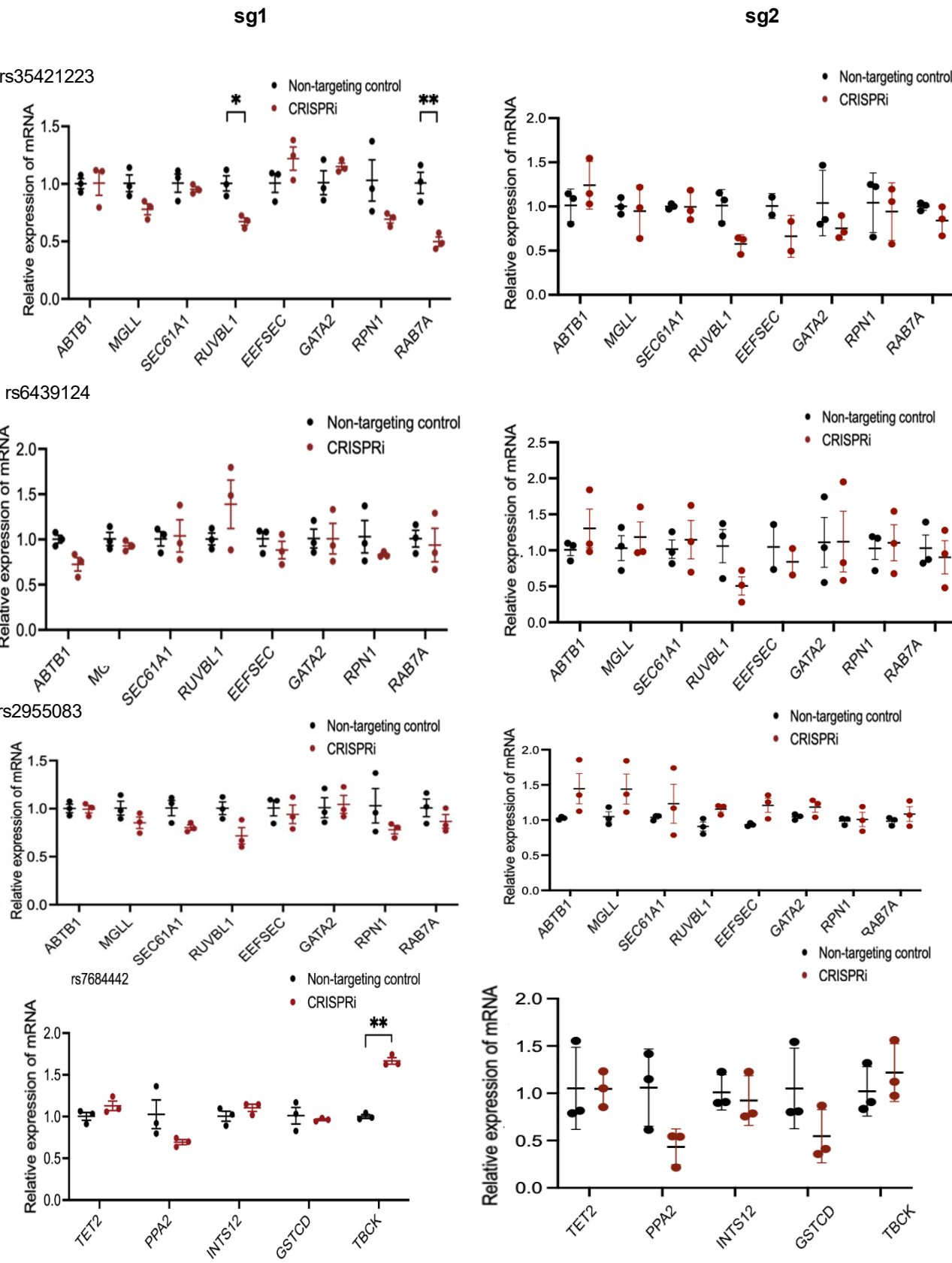


Figure S7. Protein-level validation of CRISPRi targeting of the rs35421223-containing regulatory element in NHBE cells.

